Artificial Intelligence Driven Meta-Analysis

Introduction to Meta Analysis



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Reminder for Group Project

1. **Date of Final Report：** Dec. 5, 2025
   - **Deadline of Submission：** 10.00 am
2. **Date of Presentation：** Dec. 12, 2025
   - **Time：** 10:00-13:00 and 19:00-22:00
   - **Deadline of Submission：** 10.00 am



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Major Requirements for Group Project

- Teams consist of **3 or 4 randomly** assigned members
- The project data will be released on **Sep 12, 2026** (today)
- Your team need to select **a topic** and utilize it to develop a detailed project.
- Teams are required to submit ***the final project reports with a Turnitin report, the ppt or pdf of the presentation slides, the work contribution and Gen AI documents***.
- Teams should select their **own presenter**.
- The project leader will submit **all project files**, and every member must submit a **peer evaluation for the entire team.**
- **For more details, please refer to the files in Canvas.**

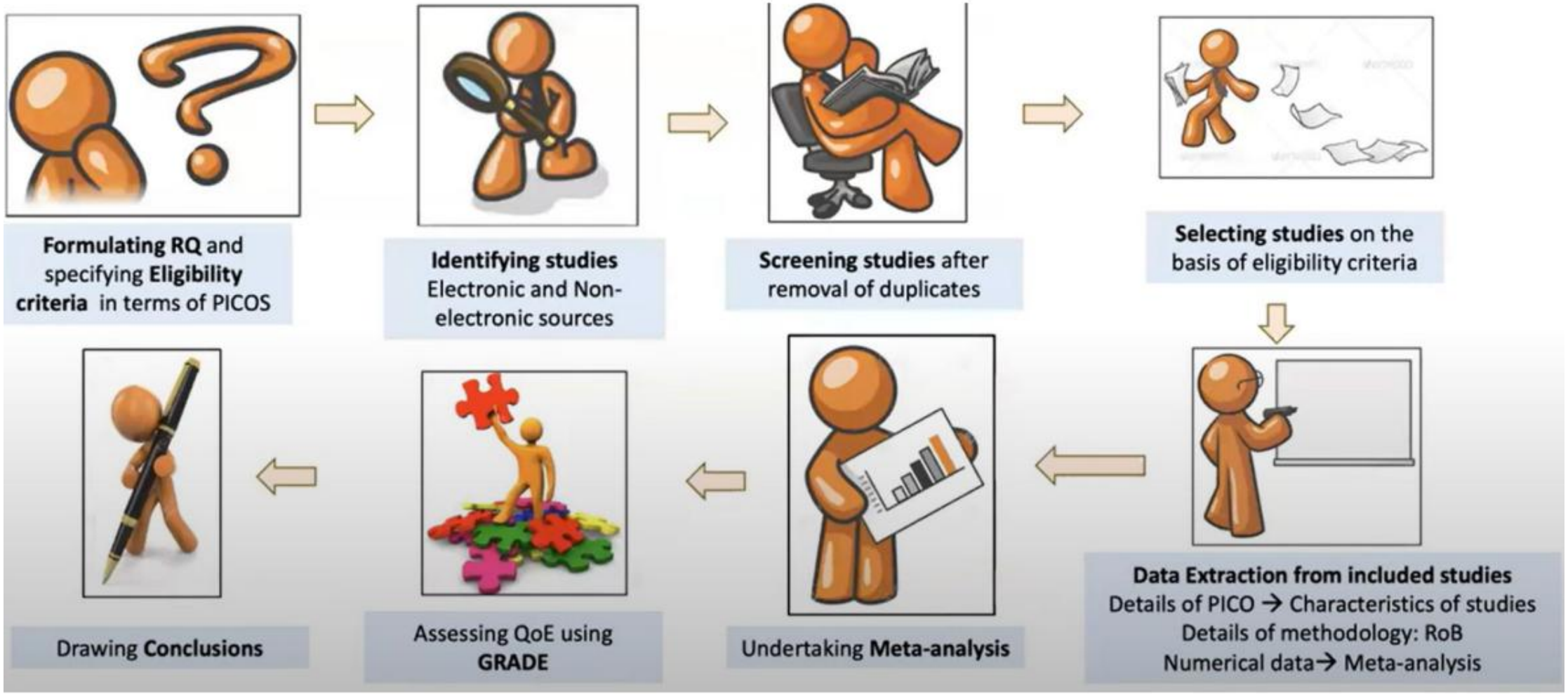


Systematic Review And Meta-Analysis

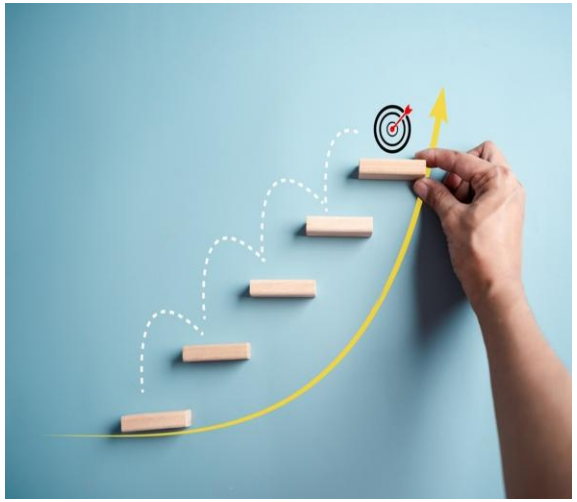
A Step by Step guide Approach


Garbage in
Garbage OUT

Key Steps in a Meta-Analysis



Key Steps



1.Defining the review question

2.Searching for studies (Search Strategy)

3. Screening studies

4.Selecting the studies

5.Data extraction

6. Data analysis (Meta-analysis)

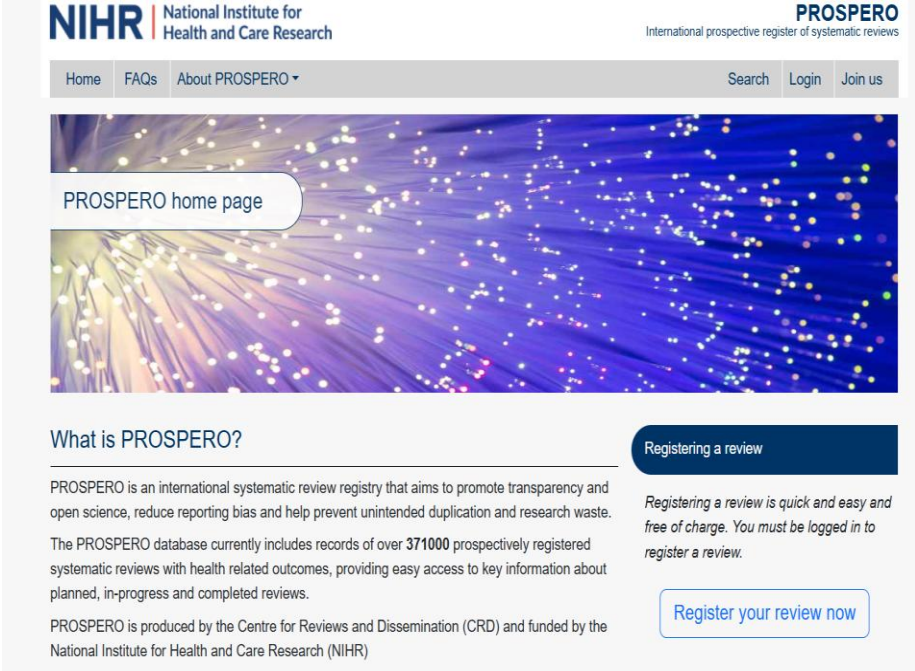
7. Reporting

8. Updating

SRMA- General points to be considered

- **Team- at least 3 members**
  - Content/ Subject experts
  - Research/ Review experts
  - Statistician
- **Prospectively register protocol –**

  PROSPERO。。。
- **Robust search strategy**
- **At least 3 databases**



RESEARCH TOPIC IDENTIFICATION

Steps of choosing research topic:

1. Brainstorming
2. Read background information.
3. Specify your topic (not too broad, not too narrow).
4. Formulate your focused research topic title.



The Best Way to Identify Meta-Analysis Research Topics

Supervisor tell you a research topic!



Daily clinical practices/
Research work



STEP 1: BRAINSTORMING

1 Brainstorming to choose the correct topic ideas:

1. **Relevant:**
   Interesting to you and your specialty.
2. **Feasible:**
   All required resources, funds and participants are available and can be achieved within time-frame.
3. **Nobility:**
   The topic should not be researched too much previously.
4. **Ethical:**
   Do not cause harm to you or participants.



STEP 2: READING

2 Reading background information:

- Read a general information on the topic.
- Use current articles on the topic and online data-base.
- Ask help from librarian if you face any difficulty.

"This step is important prior to narrowing your topic idea into focused sub-topic."

STEP 3: SPECIFY

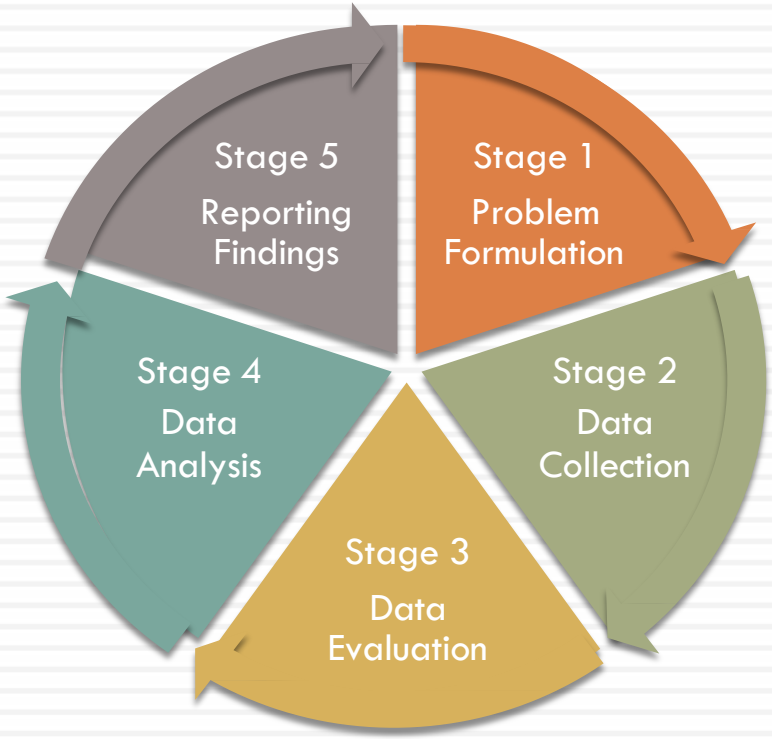
❸ Focus on and specify your topic (Sub-topic):

1. **Area of interest:**
   Specify some aspect for that topic to be studied.
2. **Geographic context:**
   Specify the region you want to conduct your research in.
3. **Time frame:**
   Specify the time frame for your research.
4. **Population group:**
   Specify certain population criteria.

STEP 4: FORMULATE THE TOPIC TITLE

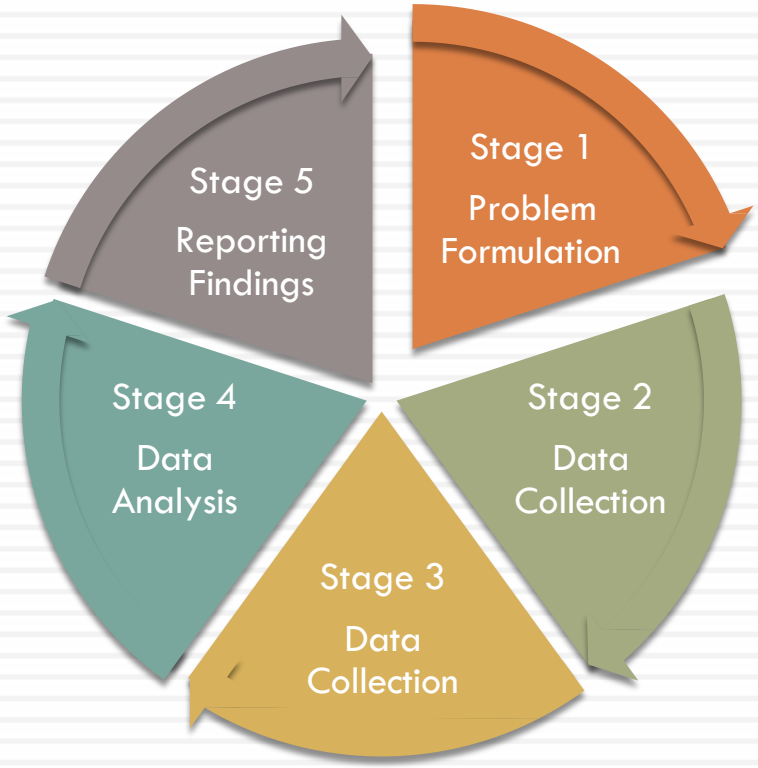
WHAT	Things to identify in topic of interest
WHO	Population
WHERE	Place, Region, Country
WHEN	Time-frame
HOW	Type of study (design)

Stages of Systematic Review Process



Campbell Collaboration

Stage 1: Problem Formulation





Problem Formulation

- Developing your protocol (research plan)
  - Developing research questions
  - Developing inclusion / exclusion criteria
  - Developing a coding manual for data extraction
  - Determining data management

Research Questions

- Address the variety of issues: incidence/ prevalence/ etiology of diseases / diagnostic accuracy/ effectiveness of interventions.
- Should be precise.
- Should specify key components: PICO

A description of the populations	**P**
An identified intervention	**I**
An explicit comparison	**C**
Relevant outcomes	**O**



Research Questions

- Questions about intervention effects
  - Are drug courts effective in reducing drug crime among adolescents, relative to practice as usual?
- Questions about associations
  - What is the relationship between teacher education/training and student achievement? Does this relationship differ for elementary and secondary school students?
- Diagnostic/Prognostic questions
  - Which risk assessment (A vs. B) is a better predictor of child maltreatment?



Research Questions

- Questions often use the PICO(S) framework:
  - **P**opulation/Participants
  - **I**nterventions
  - **C**omparisons
  - **O**utcomes
  - **S**tudy design

- "To assess the effects of [*intervention*] compared to [*comparison/control*] for [*condition/problem*] in [*population*] in [*context*] on [*outcomes*]."



Research Questions

1.In older adults(P), is an educational intervention(I) more effective than no intervention or an alternative intervention(C) in

1. improving driving awareness/knowledge,
2. improving on-road driving behavior and skills, and,
3. reducing crash rates? (O)



Research Questions

2. In post-cardiac arrest patients with return of spontaneous circulation **(P)**, does therapeutic hypothermia **(I)** compared with usual care **(C)**, improve morbidity or mortality **(O)**?

- Developing a research question for a systematic review requires an understanding of the existing literature, including gaps and uncertainties, definitions, and terminologies

Research Questions

- Research question
  - Is drug therapy associated with long-term morbidity and mortality in older persons with moderate hypertension?



Research Questions

- Research question
  - Is drug therapy associated with long-term morbidity and mortality in older persons with moderate hypertension?
- P = Older persons with moderate hypertension
- I = Drug therapy
- C = Not stated (presumably any intervention other than the named drug therapy)
- O = Long-term morbidity and mortality

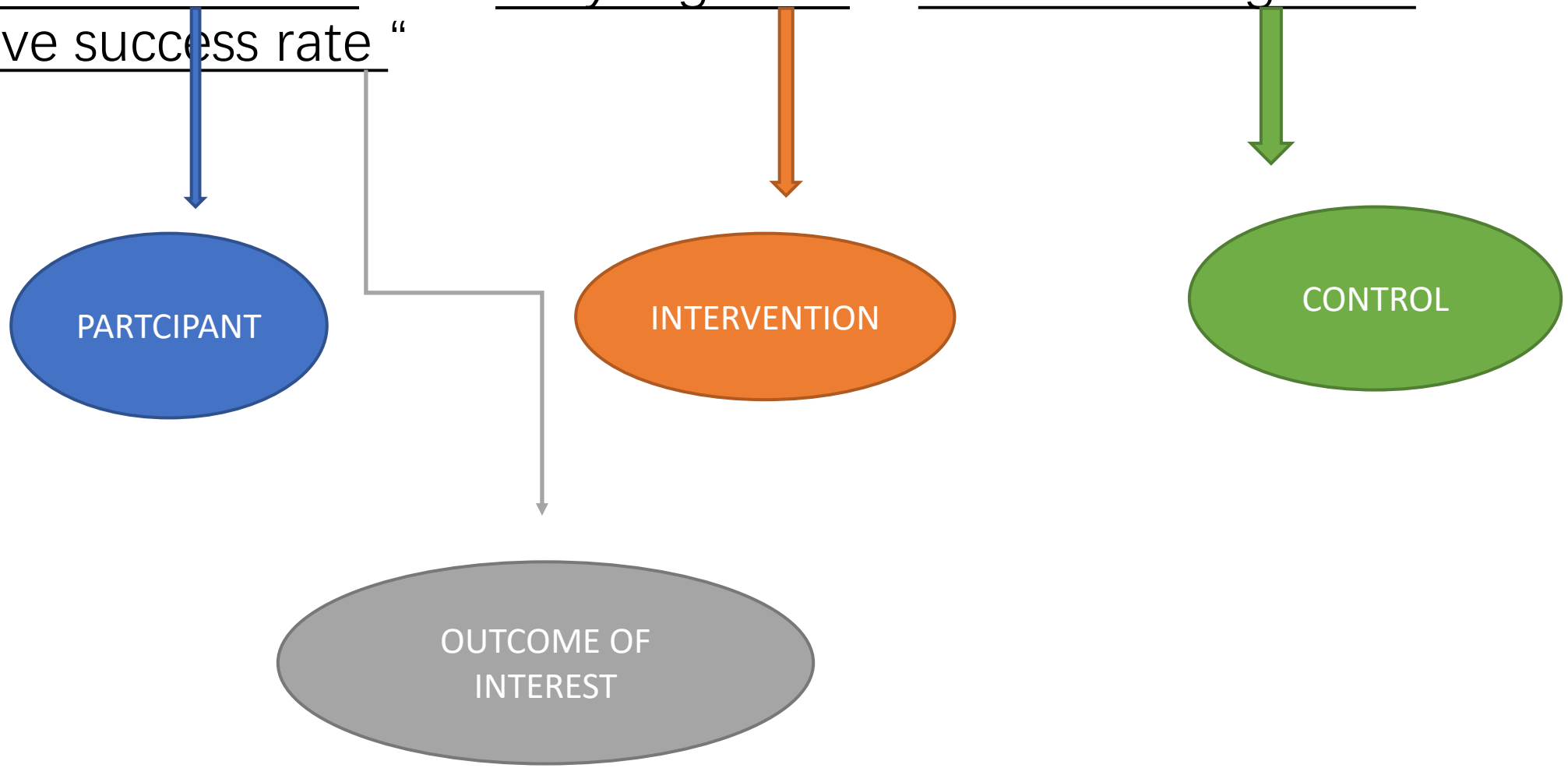
Research Questions

" in patients with TB does daily regimen or alternative regimen have effective success rate "



Research Questions

" in patients with TB does daily regimen or alternative regimen have effective success rate "





Inclusion / Exclusion Criteria

- Study eligibility criteria:
  - Clearly defines which *subjects* or studies will be included in the review
    - Essential for determining relevant and irrelevant studies
  - Determined by the research question
  - MUST be defined before data collection (i.e. conducting the search)

Inclusion / Exclusion Criteria

Criteria is set up with relevance to PICO components

P= PARTCIPANTS → Sociodemographic characteristics and study setting

I=INTERVENTIONS → What intervention? how delivered? Who delivers?
Intensity of intervention?

C=CONTROL → What kind of comparison?
Active control : different regimen of same drugs/ diff therapy
Inactive control: placebo/ standard protocol/ no Rx

O = OUTCOME → Should be clearly stated.
Type of outcomes: short term or long term
Primary or secondary outcome
One benefit and one ADR should also be assessed.



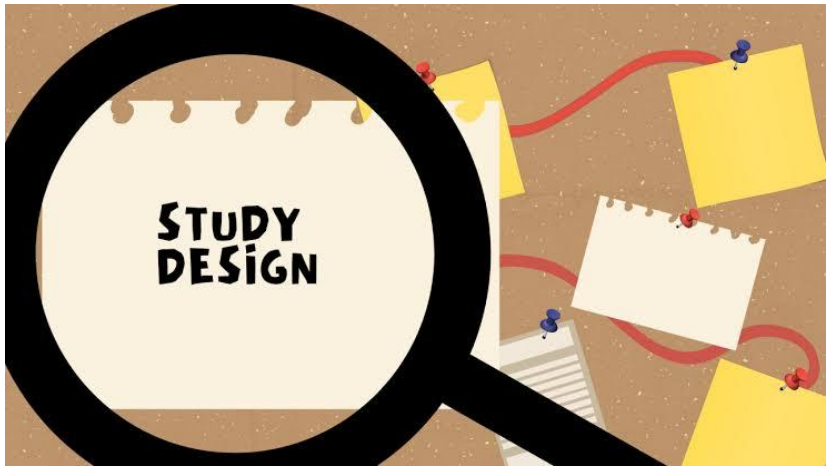
Inclusion / Exclusion Criteria

- Criteria to consider:
  - Population/ Participants:
    - Age / grade level
    - Gender, ethnicity, socioeconomic status
    - Special or clinical populations
  - Interventions
    - What types of interventions are relevant
    - Identify elements of a program/intervention that are critical
  - Outcomes
    - What is the purpose of the intervention (e.g. quality of life, transition, etc)
  - Study Design
    - What study designs are most appropriate (e.g. RCT)

Inclusion / Exclusion Criteria

- **Define the study design also based on the research question**

1. Prevalence of diseases/ diagnostic accuracy of tests: **cross-sectional design**
2. Aetiology of disease : **cohort design**
3. Effects of intervention: **RCT**





Inclusion / Exclusion Criteria

- Criteria to consider:
  - Geographic considerations
  - Time restrictions
    - If limiting the research to specific dates, indicate reason
    - Example: interventions for low birth weight babies – date limit was 1992 because of surfactant trials
  - Language restrictions
    - Remember, it can be difficult to obtain articles in Portuguese even if you are fluent

Inclusion / Exclusion Criteria

Objective: Investigate the effectiveness of peer-mediated interventions (PMI) in the treatment of individuals with autism spectrum disorders (ASD)

1.2. Inclusion and exclusion criteria

Articles were included in this review if a PMI was used to treat a person diagnosed Autistic Disorder, Asperger's Syndrome, or Pervasive Developmental Disorder Not Otherwise Specified (PDD-NOS). Studies were excluded if they did not explicitly state the participants' diagnosis or if the diagnosis did not clearly fit the diagnostic criteria for Autistic Disorder, Asperger's Syndrome, or Pervasive Developmental Disorder Not Otherwise Specified (PDD-NOS) (American Psychiatric Association, 2000). Additionally, studies were excluded if peers played too limited of a role in the intervention. Specifically, this excluded two types of involvement: (a) peers were only used as recipients of social initiations without any training, and (b) peer modeling, in which target children observed peers perform particular behaviors without any interaction between the two. A total of 42 articles met these criteria and are included in this review.

Inclusion / Exclusion Criteria

Objective: To determine the effectiveness of mindfulness-based interventions with people diagnosed with multiple sclerosis

Table 1 SPIO narrow screen inclusion/exclusion criteria

	Inclusion criteria	Exclusion criteria
Study design	Randomised controlled trial, controlled trial	Qualitative studies Single case study Systematic reviews Literature reviews Guidelines Audit
Population	Age >18 years Any diagnosis of MS	<18 years old Diseases other than (and not including) MS
Intervention	Any specifically mindfulness-based intervention (MBI)	Psychotherapy Drug treatments Manual therapy (ie massage)
Outcomes	Perceived stress Anxiety Depression HRQOL Pain Personal wellbeing Social participation	

Note: SPIO is similar to PICO

Inclusion / Exclusion Criteria

Systematic Review of CVD Risk Assessment Tools

Abstract Review Form

First Author, Year: _____ Ref ID #_____ Abstractor Initials: ___ ___ ___

Primary Inclusion/Exclusion Criteria			
1. Relevant to SER topic	Yes	No	Cannot Determine
2. Attempts internal or external validation	Yes	No	Cannot Determine
3. Published in English	Yes	No	Cannot Determine
4. Original research (exclude reviews, editorials, commentaries, letters to editor, etc.)	Yes	No	Cannot Determine
5. Eligible Study types a. Randomized controlled trials b. Controlled clinical trials c. Cohorts d. Case-series	Yes	No	Cannot Determine
6. Reports relevant outcomes: a. Coronary Artery Disease (non fatal and fatal MI and sudden coronary heart disease death) b. Cerebrovascular stroke (thrombotic/hemorrhagic) c. Combination of a and b	Yes	No	Cannot Determine
7. Study size > 200 If No, state study size_____	Yes	No	Cannot Determine
8. Study population is adults asymptomatic for CVD If No, state % symptomatic _____%	Yes	No	Cannot Determine

Retain for:

_____BACKGROUND/DISCUSSION

_____REVIEW OF REFERENCES

_____Other_____



Inclusion / Exclusion Criteria

Physical exercise and individuals with autism spectrum disorders (ASD):

- Objective: review of interventions designed to increase the exercise behavior of individuals with ASD

Inclusion Criteria	Exclusion Criteria



Inclusion / Exclusion Criteria

Physical exercise and individuals with autism spectrum disorders (ASD):

- Objective: review of interventions designed to increase the exercise behavior of individuals with ASD

> The studies were selected on the basis of the following criteria: (i) research published between 2003 and 2024; (ii) pilot and experimental studies (randomized clinical trials or single subject designs), published in peer-reviewed journals; (iii) studies with a sample of children, young people and/or adults diagnosed with ASD (autistic disorder; Asperger's syndrome and pervasive developmental disorder without other specification), in different le vels; (iv) no restrictions on ethnicity, gender, age group, sample number or duration of the physical exercise program; (v) studies that used cardiorespiratory fitness and/or muscular strength as the dependent variable; (vi) studies that have assessed cardiorespiratory fitness and/or muscle strength; and (vii) studies that have implemented a group physical exercise program as an independent variable.



Inclusion / Exclusion Criteria

Physical exercise and individuals with autism spectrum disorders (ASD):

- Objective: review of interventions designed to increase the exercise behavior of individuals with ASD

> The studies were excluded on the basis of the following criteria: (i) articles published before 2003; (ii) articles not published in English, Spanish or Portuguese; (iii) systematic review, meta-analysis and/or thesis/dissertation articles; (iv) studies that covered other types of disability without ASD; (v) studies that did not use cardiorespiratory fitness and/or muscle strength as a dependent variable; (vi) studies that did not assess cardiorespiratory fitness and/or muscle strength; and (vii) studies that did not describe the intervention protocol, namely, the physical exercise program.



Coding Manual

Data extraction

- A coding manual to extract data (i.e. study details) from each study needs to be developed prior to collecting data (i.e. searching)
- Will be discussed under Data Extraction section



Data Management

PRISMA – Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

- It is an evidence-based minimum set of items for reporting in systematic reviews and meta-analyses.
- The aim of the PRISMA Statement is to help authors improve the reporting of systematic reviews.
- The PRISMA Statement consists of a 27-item checklist and a four-phase flow diagram.

PRISMA 27-item checklist

- http://www.prisma-statement.org/statement.htm

Data Management

Bibliographic Management

- Determine at the beginning of the research planning, what bibliographic management software you will use to manage and track data (i.e. references, citations)
  - Endnote ($)
    - Library supports
  - Refworks (free)
    - Library supports
  - Mendeley (free)
    - Library supports
  - Zotero (free)
  - Covidence (official tool from Cochrane)

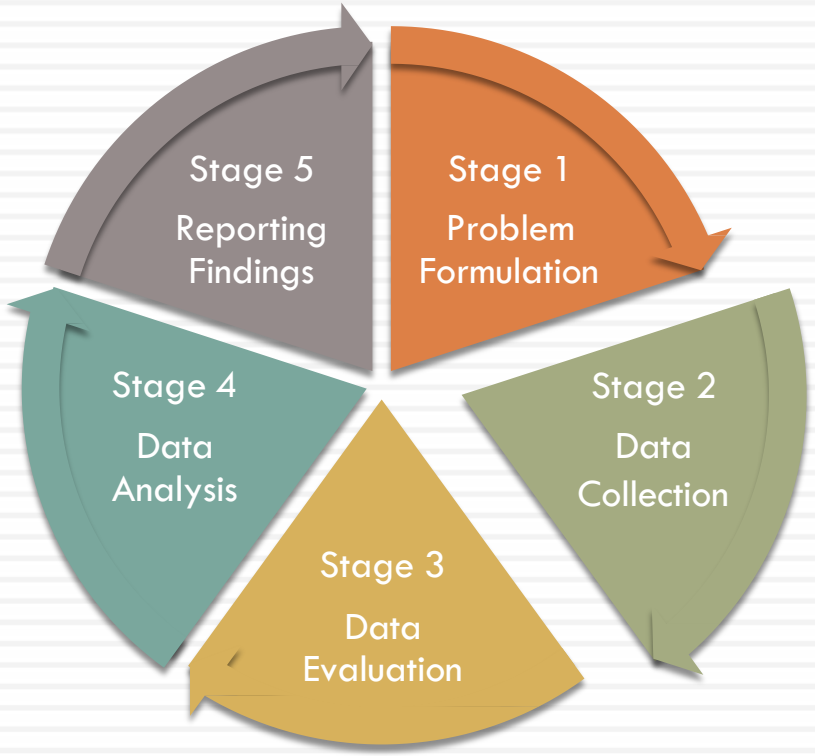


Data Management

Use bibliographic management software to:

- Maintain a searchable database of references related to the systematic review
- Store all references selected for the systematic review
- Keep track of number of references from each database
- Determine number of duplicate references
- Store all discarded references
- Create citations and bibliography when writing up the results of the SR

Stage 2: Data Collection





Steps in Data Collection

- The Literature Search
- Developing inclusion / exclusion criteria
- Identifying resources to search

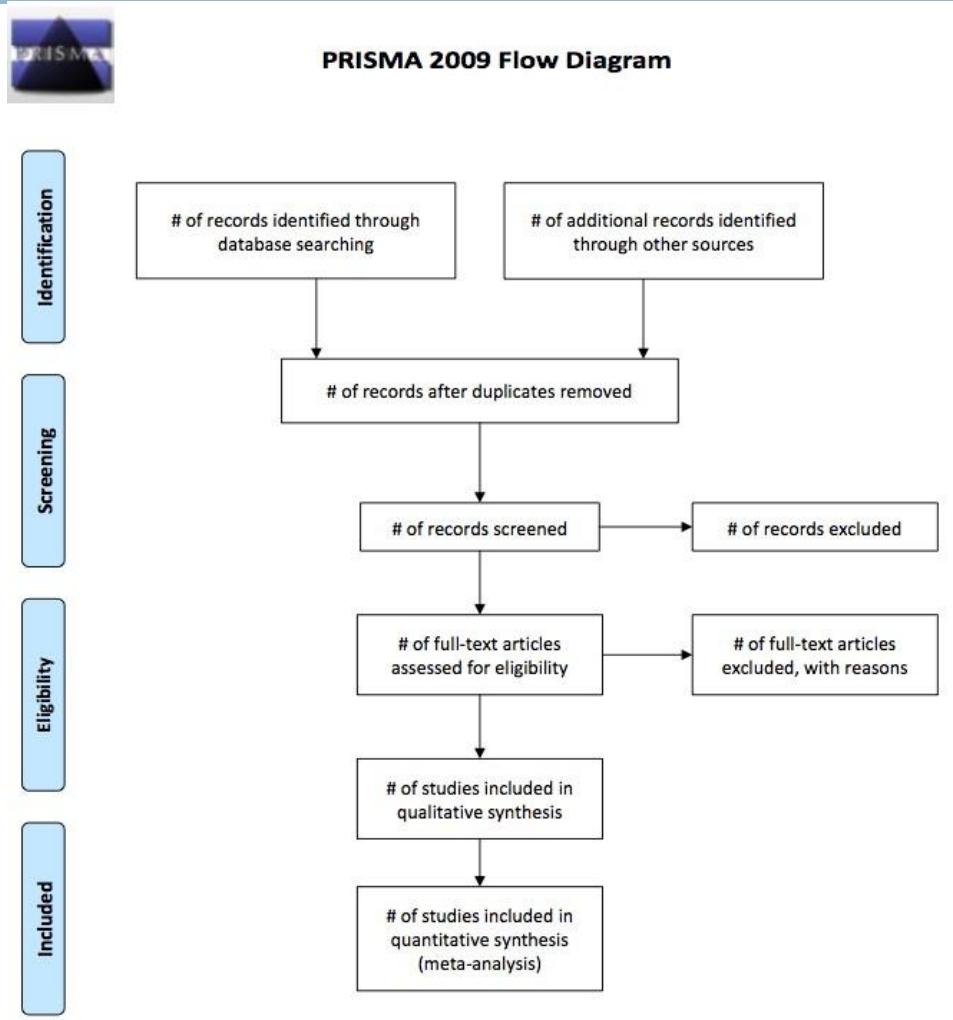


The Literature Search

- The literature search is the most important part of a systematic review
  - The literature search is how you obtain your data for your study
- The goal of the literature search is to find all relevant studies including:
  - Traditional peer reviewed literature
  - Grey literature
- The literature search must be comprehensive and systematic

The Literature Search

- The systematic review literature search must:
  - Be well documented
  - Be transparent and reproducible
  - Include a diversity of resources
  - Be iterative



From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). *P*referred *R*eporting *I*tems for *S*ystematic Reviews and *M*eta-*A*nalyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097



The Literature Search

- Consult with your subject librarian
- Document Document Document
  - Databases searched, date range searched, date last searched
  - All search terms for each database, as far as possible
  - Follow the PRISMA (or other) flowchart
    - Number of hits for each database searched
    - Duplicates within databases



The Literature Search

- Search Strategy
  - Key concepts to be searched (PICOS if using)
    - How are these terms represented in different disciplines?
    - What are related terms / synonyms? Be as comprehensive as possible
    - What are the subject headings (e.g. MeSH) in each database?
  - Review known relevant articles to determine keywords and subject headings
  - Review previous relevant systematic reviews for possible strategy
    - Is there a Cochrane group that has recommended terms

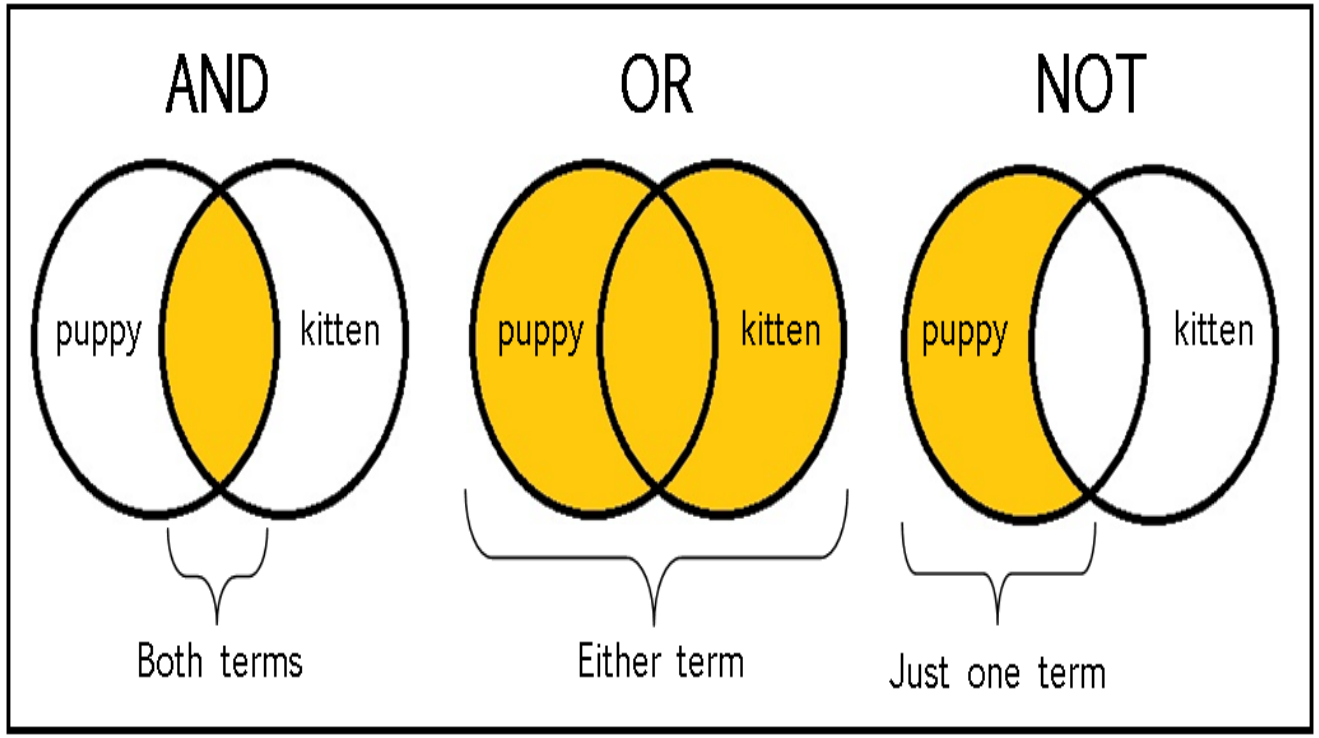


The Literature Search

- Search Strategy
  - Use truncation
    - nurs* will retrieve: nursing, nurse, nurses, nursed, etc
  - Use both free text (keywords) and subject headings (i.e. controlled vocabulary), when possible
  - Pilot your search strategy
    - Is your search retrieving the known articles?
    - If no, why not?
      - Check indexing
      - Check truncation
      - Check free text keywords
      - Check descriptors / subject headings
  - It takes a LOT OF TIME

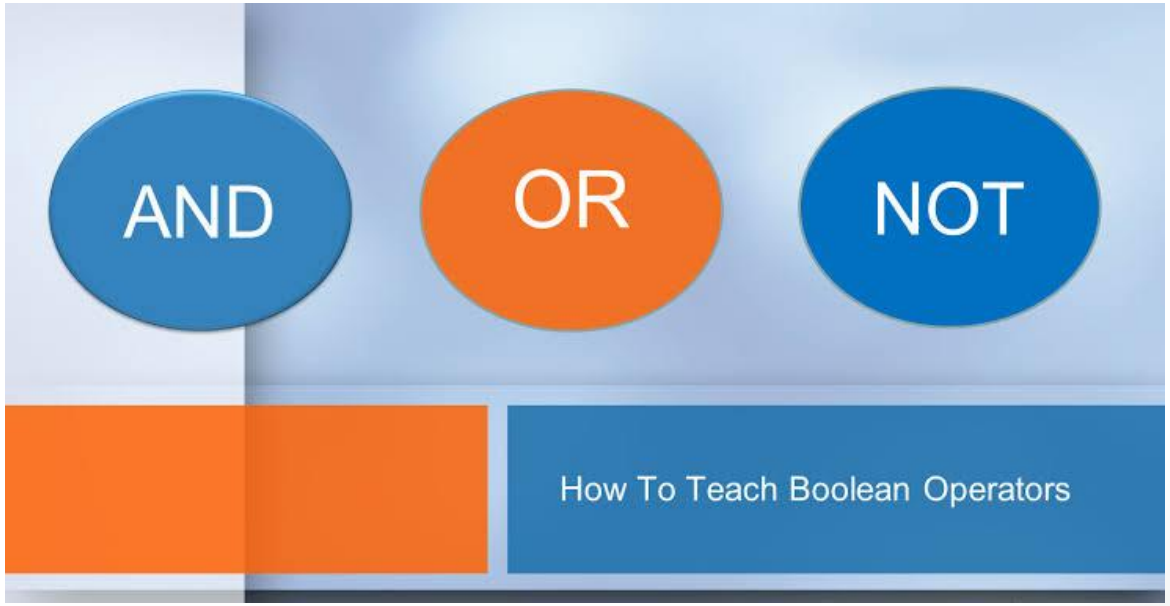
How to do the search?

- Each database should be thoroughly searched
- MESH/ key words can be used
- Use the Boolean operators



Boolean operators

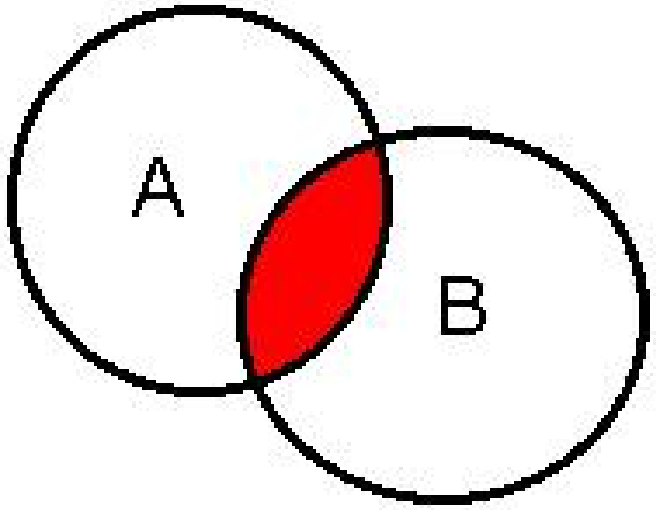
- Use **BOOLEAN OPERATORS** to relate the terms in specific ways that will affect the results of a search.
- Boolean operators are terms that create relationships between concepts and words
  - **AND**
  - **OR**
  - **NOT**



Boolean operators

AND

retrieves only articles that contain <u>all</u> the terms and will **narrow** the search



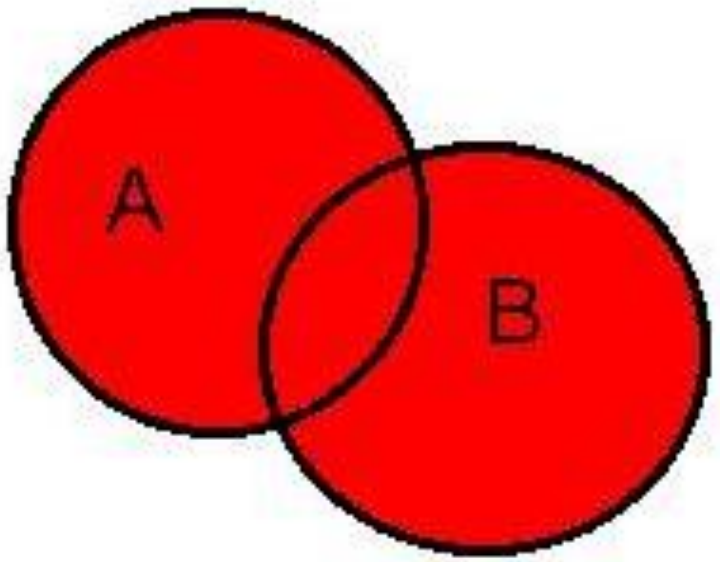
Examples:

- “child obesity” **AND** Maryland
- “child obesity” **AND** “eating disorder”
- “video games” **AND** children
- “attention deficit disorder” **AND** children

Boolean operators

OR

retrieves articles with <u>any</u> of the terms and will **broaden** the search



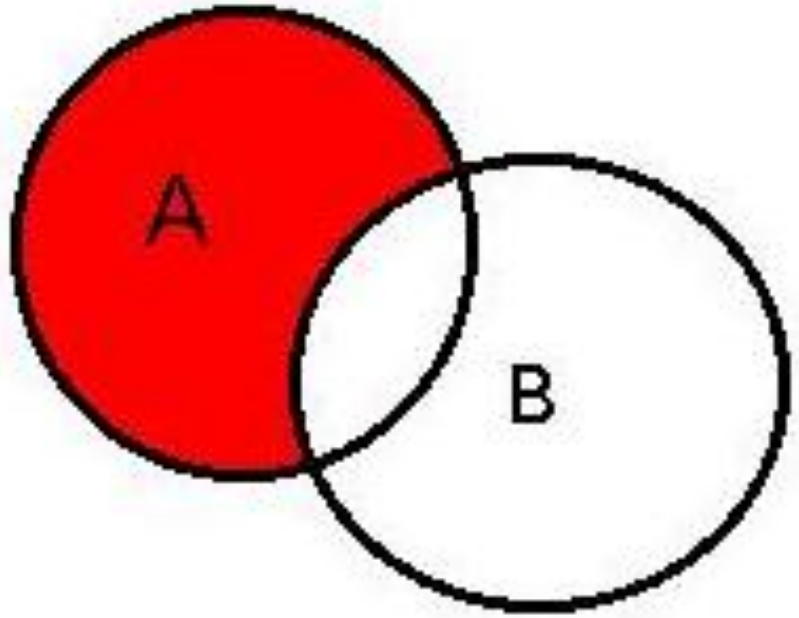
Examples:

- “child obesity” **OR** “overweight children”
- children **OR** juveniles
- “attention deficit hyperactive disorder” **OR** hyperactivity

Boolean operators

NOT

eliminates articles containing the second term even when the first term is present. This will **narrow** the search

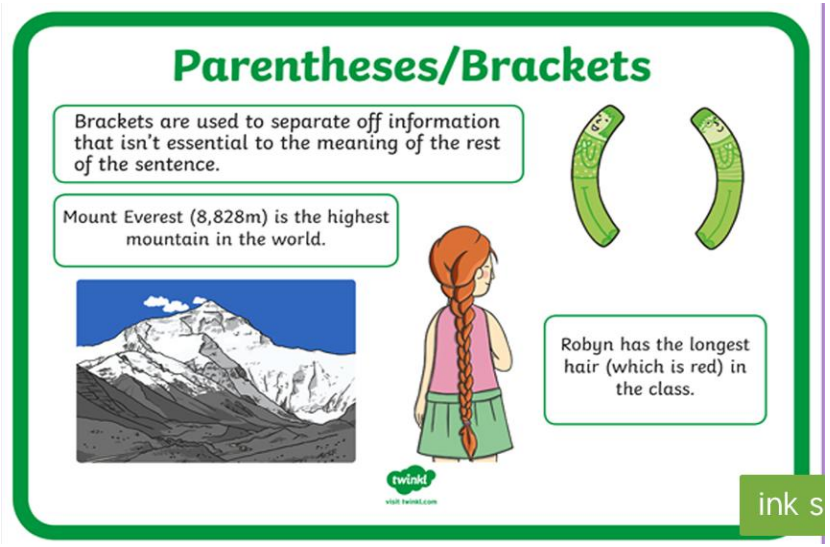


Examples:

- obesity **AND** children **NOT** adults
- depression **AND** teen **NOT** adults
- “video games” **AND** teenagers **NOT** children

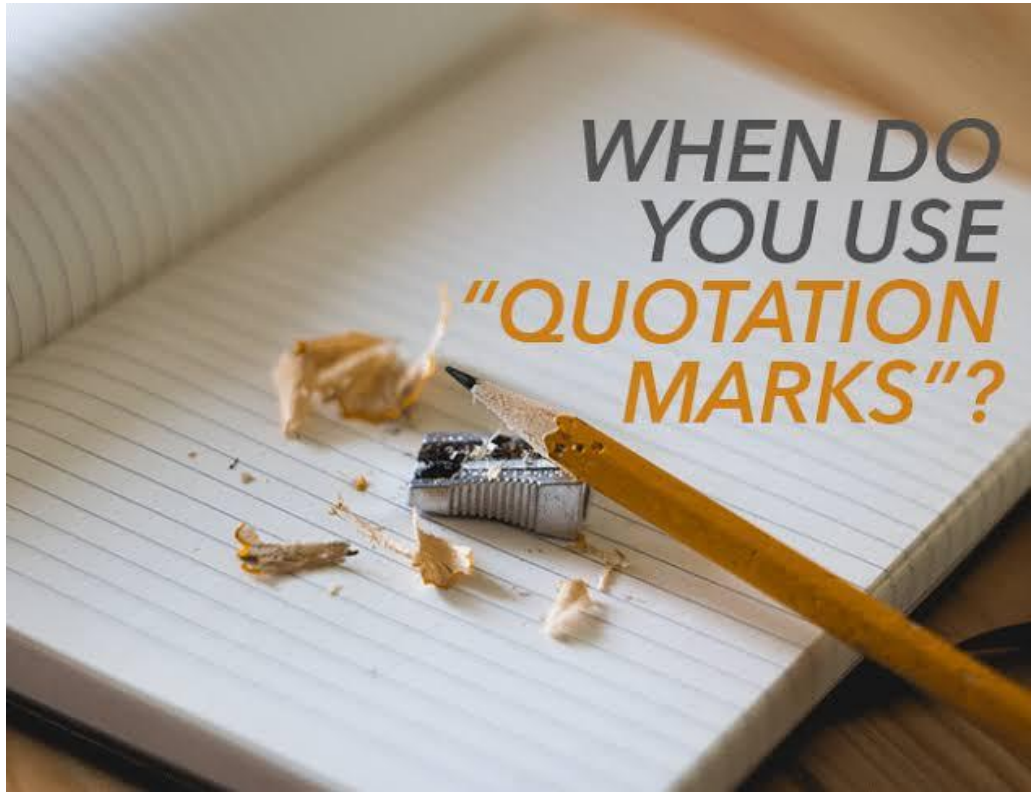
Using Nesting with Operators

- Use nesting to build a more complex search by putting keywords and/or **phrases in parentheses** to determine their relationship when more than one operator is used.
- Examples:
  - (children **AND** teenagers) **AND** (ADHD **OR** hyperactivity)
  - (children **AND** “video games”) **NOT** (teens **OR** adults)



Key Concepts

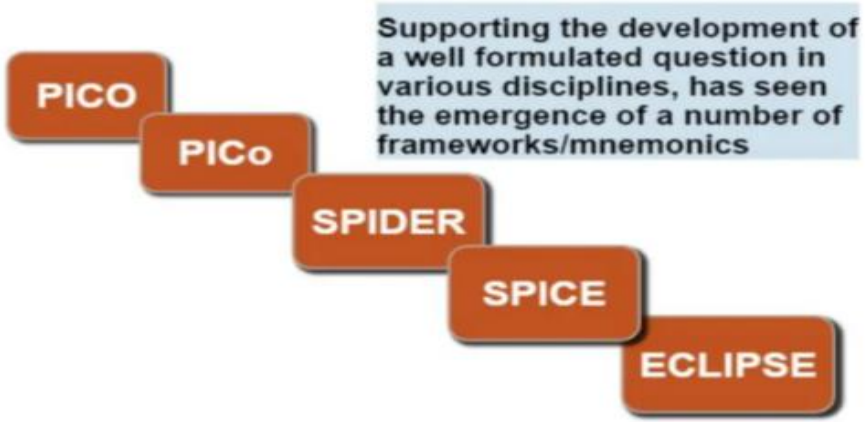
- USE **QUOTATION MARKS** " " before and after a phrase, two or more words that must be together in a defined order.
  - Examples:
    - “African American”
    - “eating disorder”
    - “obsessive compulsive disorder”
    - “video games”
- This lets the database know that you want to search for this phrase, and not the individual words that make up the phrase.



Structuring your search strategy using the PICO/ST framework

- Frameworks are used to break up the research topic into key concepts & in the developing of research questions
- There are various frameworks/mnemonics that can assist you break up the research topic into key concepts

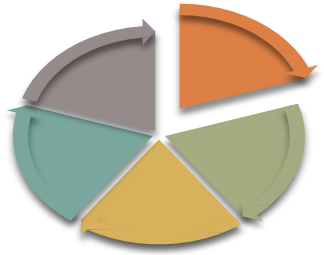




Frameworks

- PICO
- PICo
- SPIDER
- SPICE
- ECLIPSE

Frameworks/mnemonics to support development of a well formulated question in various disciplines



Frameworks

- **PICO**: Population, Intervention, Comparison, Outcome (PICOT, PIO, PICOS) - reviews of interventions for health, appropriate for quantitative studies
- **PICo**: Population/Problem, Phenomenon of Interest, Context – qualitative studies
- **SPIDER**: Sample – Who; Phenomenon of Interest – What; Design – How; Evaluation – Compared with; Research type – Which types – qualitative research
- **SPICE**: Setting –Where; Perspective – For Whom; Intervention – What else; Comparison – Compared with; Evaluation – With what result? - social sciences

Subject Headings

What is MeSH?

MeSH = Medical Subject Headings

MeSH is a controlled vocabulary used by databases (like PubMed!) to categorize articles by topic!

Database	Subject Headings
Medline/PubMed	MeSH
EMBASE	EMTREE
CINAHL	CINAHL Headings
Cochrane Library	MeSH
PsycINFO	Thesaurus of Psychological Index Terms
Scopus, Web of Science	N/A

Tips:

Select subject headings that are the closest match for your concepts

Pay attention to “explode” commands – PubMed will search related headings by default

Which Subject Heading Explodes?

A. Bird Diseases

All MeSH Categories
- Diseases Category
  - Animal Diseases
    - **Bird Diseases**
      - Avian Leukosis
      - Fowlpox
      - Influenza in Birds
      - Malaria, Avian
      - Marek Disease
      - Newcastle Disease
      - Poultry Diseases
        - Enteritis, Transmissible, of Turkeys
        - Influenza in Birds
        - Poult Enteritis Mortality Syndrome
      - Reticuloendotheliosis, Avian
      - Sarcoma, Avian
      - Tuberculosis, Avian

B. Neutropenia

All MeSH Categories
- Diseases Category
  - Hemic and Lymphatic Diseases
    - Hematologic Diseases
      - Leukocyte Disorders
        - Leukopenia
          - Agranulocytosis
            - **Neutropenia**

Identify Subject Headings

What are MeSH headings for these concepts?

Hydration	Delirium	End of Life

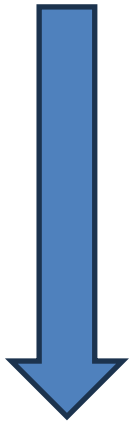
Add appropriate subject headings to your worksheet

Identify Subject Headings

What are MeSH headings for these concepts?

Hydration	Delirium	End of Life





☐ **Delirium**

1. A disorder characterized by CONFUSION; inattentiveness; disorientation; ILLUSIONS; HALLUCINATIONS; agitation; and in some instances autonomic nervous system overactivity. It may result from toxic/metabolic conditions or structural brain lesions. (From Adams et al., Principles of Neurology, 6th ed, pp411-2)

Death

Irreversible cessation of all bodily functions, manifested by absence of spontaneous breathing and total loss of cardiovascular and cerebral functions.

PubMed search builder options

☐ Restrict to MeSH Major Topic.
☐ Do not include MeSH terms found below this term in the MeSH hierarchy.

Tree Number(s): C23.550.260
MeSH Unique ID: D003643
Entry Terms:

- End Of Life
- End-Of-Life
- Determination of Death
- Near-Death Experience
- Cardiac Death
- Death, Cardiac

Next: Identify Text Words

Why?

1. Not all citations have subject headings

Indexers are only cyborgs



Image credit: http://cdn.zmescience.com/wp-content/uploads/2011/09/cyborg.jpg

How to Search

- Start with your first concept
  - Search for the subject headings first
  - Then search text words
  - Combine these synonymous searches with OR using your search history
- Repeat for your second, third, and subsequent concepts
- Finally, combine large search results set with AND

Tip: Test your search strategy by checking to see if a few “target articles” appear in the results

How to Search

Concept 1

Search #1 =

Search #2 =

Search #3 =

Search #4 =

Concept 2

Search #6 =

Search #7 =

Search #8 =

Search #9 =



Search #5 = #1 OR #2 OR #3 OR #4

Search #10 = #6 OR #7 OR #8 OR #9

Search #11 = #5 AND #10

Results



The Literature Search

Are SSRIs effective for treating autism in children?

SSRIs AN	D Autism AN	D Children
Selective serotonin reuptake inhibitor* OR Serotonin update Inhibitor OR 5-ht update inhibitor* OR SSRI	Autis* OR Infantile autism OR Autistic disorder OR ASD OR Asperger*	Child* OR Infan* OR Todder* OR Preschool*



The Literature Search

- Save your searches in each database
  - Create an account within the database interface
  - Label the search clearly (is it a draft, final, date, database name)
- Keep detailed records of all searches conducted:
  - Month, date, year of search
  - Database searched
  - Exact search strategy
  - Any limits used (year of publication, language, etc)
  - Number of hits for each database
- Remember – goal is to report search strategy in a transparent manner so that it can be replicated



The Literature Search

- PRESS (Peer Review of Electronic Search Strategies)
  - **Translation:** Is the search question translated well into search concepts?
  - **Operators:** Are there any mistakes in the use of Boolean or proximity operators (e.g. ADJ)
  - **Subject Headings:** are any important subject headings missing or have irrelevant ones been included?
  - **Natural language:** are any natural language (i.e. free text) terms or spelling variances missing? Is truncation used correctly?



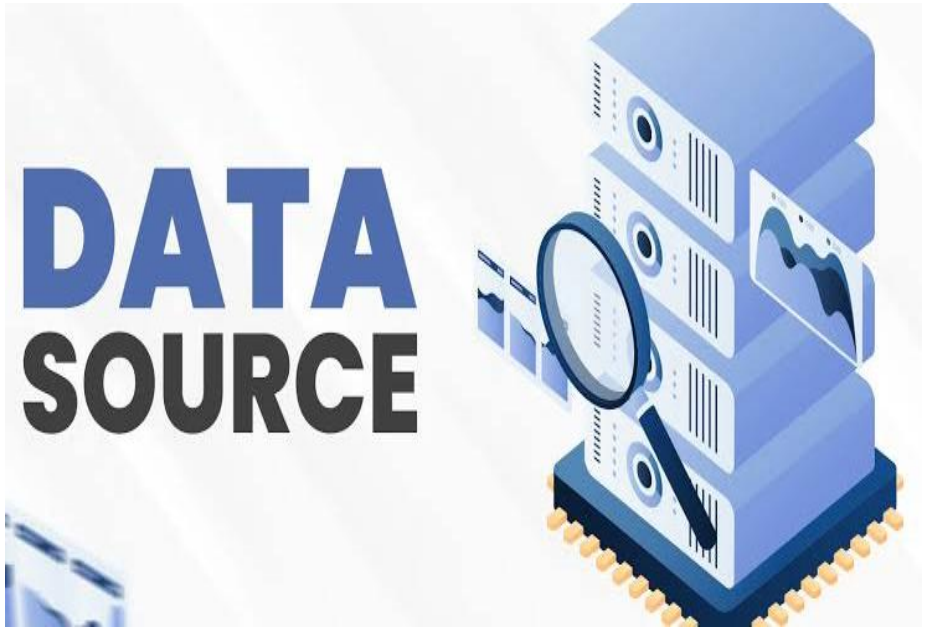
The Literature Search

- PRESS (Peer Review of Electronic Search Strategies)
  - **Spelling & Syntax:** does the search strategy have any spelling mistakes, system syntax errors, or wrong line numbers?
  - **Limits:** Do any of the limits used seem unnecessary or inappropriately used?
  - **Adapted for database:** has the search strategy been adapted for each database to be searched?
- It is a good idea to use PRESS with other researchers and your subject liaison librarian to review your search strategy systematically

Link: https://www.cda-amc.ca/press-peer-review-electronic-search-strategies

Identifying Sources

- Data sources
  - Databases
  - Reference searching
  - Cited bys (forward citations searching, citation pearl growing)
  - Contact with experts
  - Hand searching key journals, journals with poor/no indexing
  - Ongoing studies
  - Grey literature



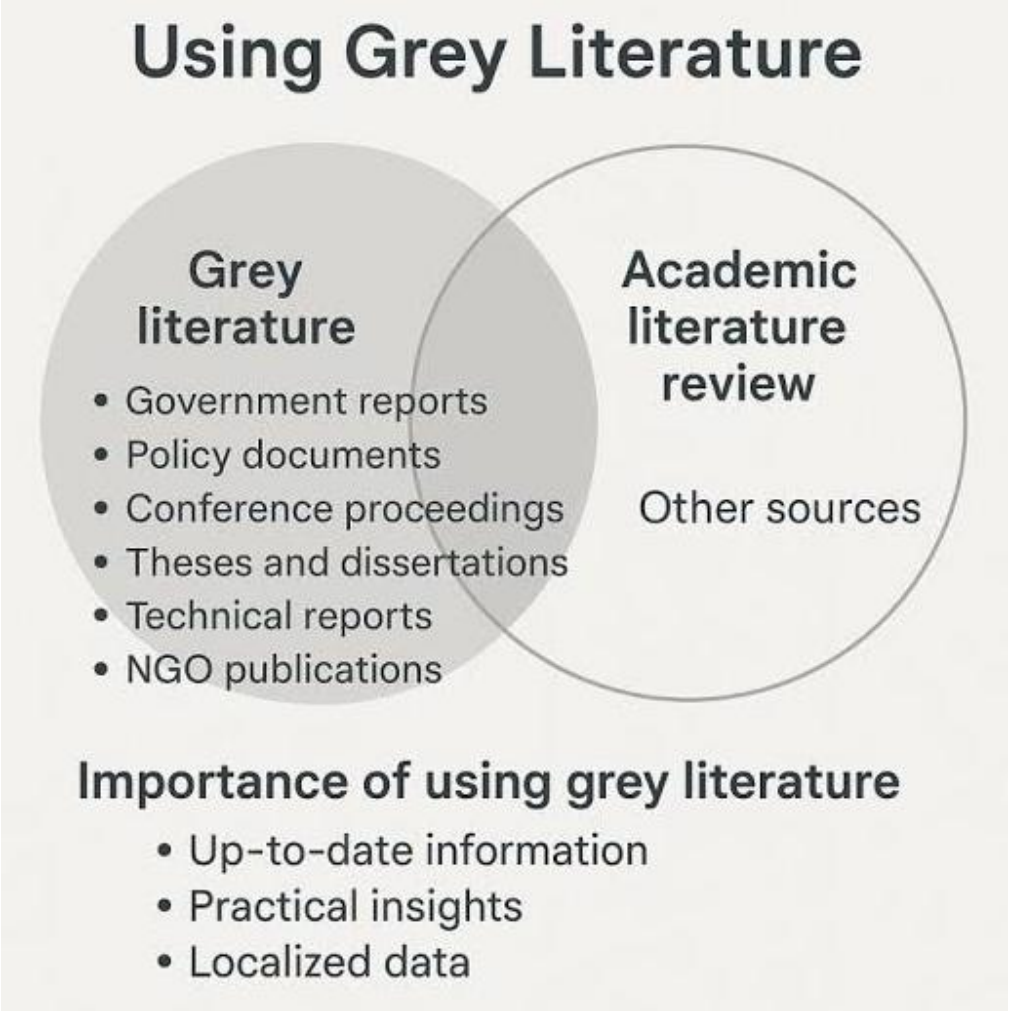


Databases

- Library databases:
  - https://library.mpu.edu.mo/zh-hant/databases
- Ask yourself: which disciplines inform my research
  - Educational interventions for young adults with ASD for gaining employment
- Different interfaces require different searching techniques
  - Review online tutorials
  - Meet with your librarian

Grey Literature

- Grey literature is defined as:
  - publicly available, foreign or domestic, open source information that is usually available only through special channels and may not enter normal channels or systems of publication, distribution, bibliographic control, or acquisition by book sellers or subscription agents (U.S. Interagency Grey Literature Working Group 1995).





Grey Literature

Grey Literature Includes:	
Government reports	Committee reports
Academic papers	Theses / dissertations
Conference papers & abstracts	Conference proceedings
Newsletters	PPT presentations
Discussion reports	Program evaluation reports
Technical documents	Standards / best practice
Working papers	. . . and more

Alberani V., Pietrangeli P. & Mazza A. (1990). The use of grey literature in health sciences: A preliminary survey. *Bulletin of the Medical Library Association, 78*, 358–363.



Grey Literature

Locating Grey Literature:

- Consider what types of organizations or associations are likely to be interested in the same research question:
  - Government agencies / NGO agencies
  - Government departments
  - Industry, trade, professional organizations
  - Advocacy groups
  - Private agencies

Grey Literature

Locating Grey Literature:

- Canadian
  - Canadian Health Research Collection
  - Canadian Public Policy Collection
- OpenGrey
  - Resource of grey literature in Europe
- Grey Literature Report
  - New York Academy of Medicine
- Dissertations / Theses
  - Proquest Dissertations and Theses
- Refer to the Systematic Review libguide for more grey literature resource

The Literature Search

Interprofessional simulation-based education (IPSE) is becoming an increasingly popular educational strategy worldwide within undergraduate healthcare curricular. The purpose of the literature review was to examine research studies featuring undergraduate IPSE.

Method

The following literature search was undertaken using the inputs of ''interprofession* OR inter discipline* OR inter-profession* OR inter-disciplin* OR multiprofession* OR multidisciplin* AND undergraduat* AND simulation''. The following databases were accessed: CINAHL, MEDLINE, and PsycINFO. Year of publication was limited from January 1999 to September 2011, and requirement of the English language was stipulated. This database search generated 120 relevant abstracts. Hand searches (including searching references from relevant articles and website searches of simulation, health care related professional/regulatory bodies) were also undertaken, see Fig. 1. Criteria were adapted from a recent review of postgraduate IPE featuring simulation (Zhang et al., 2011). The criteria used to screen all 120 abstracts included: (a) the article pertained to both simulation and undergraduate IPE and (b) the article reported a research study. Twenty two articles were considered appropriate; 4 descriptive/discussion papers were excluded and 18 articles have therefore been included in the literature review (see Table 1).

The Literature Search

Method

We conducted a systematic review of the research literature in the area of reflection and reflective learning in health professional education and practice.

Selection process

The PubMed, CINAHL, and PsychInfo databases were searched using the following keywords: *action; experience; insight; journal; personal; portfolio; professional; reflect; reflection; reflective practice; self-aware; and, self manage*. We also conducted hand searches and reviewed bibliographies of identified papers. The search was limited to English language papers published between 1995 and 2005, dealing specifically with medical or health professional education or practice.

The original search identified more than 600 papers, commentaries and reviews of the literature. To address our specific interest, we excluded all papers that did not describe research examining the process and outcomes of reflective practice in health professional education and practice. This resulted in the identification of 29 papers.

The Literature Search

The objective of this systematic review was to examine the methodological quality, measurement properties and feasibility characteristics of sporting skill outcome tests reported in the peer-reviewed literature.

2.1 Search Strategy

The literature search for this review was undertaken between July 2012 and March 2013 by the first author (SR) using the SPORTDiscus and MEDLINE databases. Key words utilised in the search using multiple combinations of AND/OR phrases included 'skill', 'measurement', 'test', 'assessment', 'reliability', 'validity', 'testing', 'elite', 'sport', 'instrument', 'sporting', 'practical', 'outcome', 'reproducibility', 'task' and 'feasibility'. Further studies were collected following examination of citations present within the collected publications ('snowballing').

The Literature Search

We therefore conducted a systematic review to determine the comparative efficacy of NPWT versus alternate TAC techniques on in-hospital mortality, length of hospital or intensive care unit (ICU) stay, fascial closure rate, and adverse events in critically ill adults with open abdominal wounds.

Search Strategy

Two surgical investigators (D.J.R., A.W.K.) created a preliminary search strategy by selecting exploded Medical Subject Heading terms and keywords for IAH/ACS, NPWT and alternate TAC techniques, trauma or sepsis, and critical care. A medical librarian (H.L.R.) subsequently refined this strategy by conducting iterative database queries and incorporating novel terms when new relevant citations were found. Searches were conducted in the following databases from their inception to July 6, 2011, without restrictions: MEDLINE, PubMed, EMBASE, Scopus, Web of Science, the Cochrane Central Register of Controlled Trials (CENTRAL), the Cochrane Database of Systematic Reviews, the Database of Abstracts of Reviews of Effects, the National Institute for Health Research Economic Evaluation Database, the Health Technology Assessment Database, and the Turning Research into Practice database (Appendix 1 shows the refined MEDLINE search). To identify unpublished studies, we also investigated two clinical trials registries (ClinicalTrials.gov and Current Controlled Trials), questioned field experts, and wrote the manufacturer of a NPWT device (Kinetic Concepts Inc. [KCI], San Antonio, TX). Additional citations were located by using the PubMed "related articles" feature and hand-searching reference lists of included trials and relevant reviews[16,26,27] and guidelines.[1,2,5] We corresponded with several authors to clarify study procedures.[17,19–21]

PRISMA Flowchart

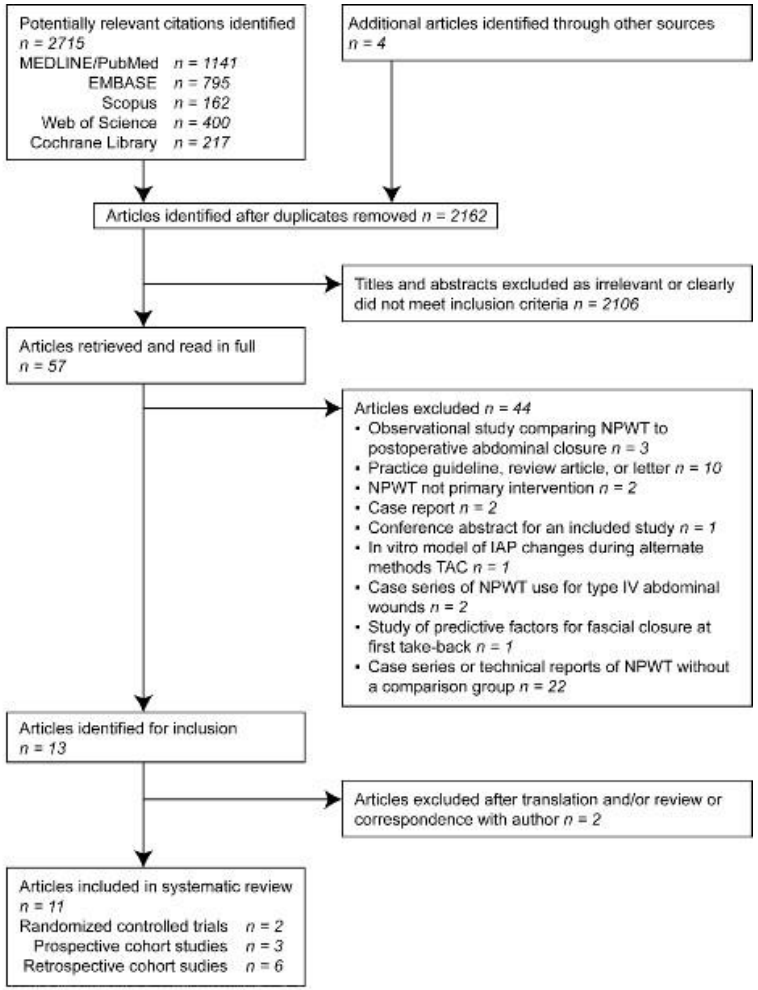
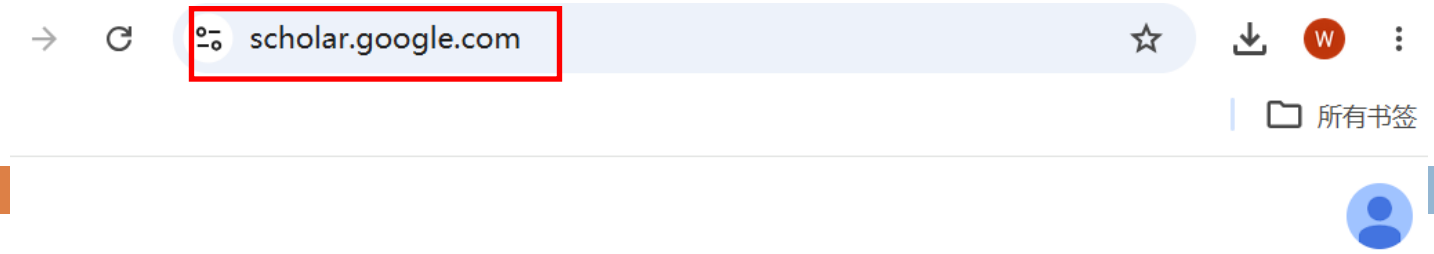


Figure 1. Flow chart of steps in systematic review.



Google Scholar

Articles Case law

New! Quick Read: see how a paper answers your question

Stand on the shoulders of giants

Google Scholar provides a simple way to broadly search for scholarly literature. You can search across many disciplines and sources: peer-reviewed papers, theses, books, abstracts and articles, from academic publishers, professional societies, preprint repositories, universities and other scholarly organisations.

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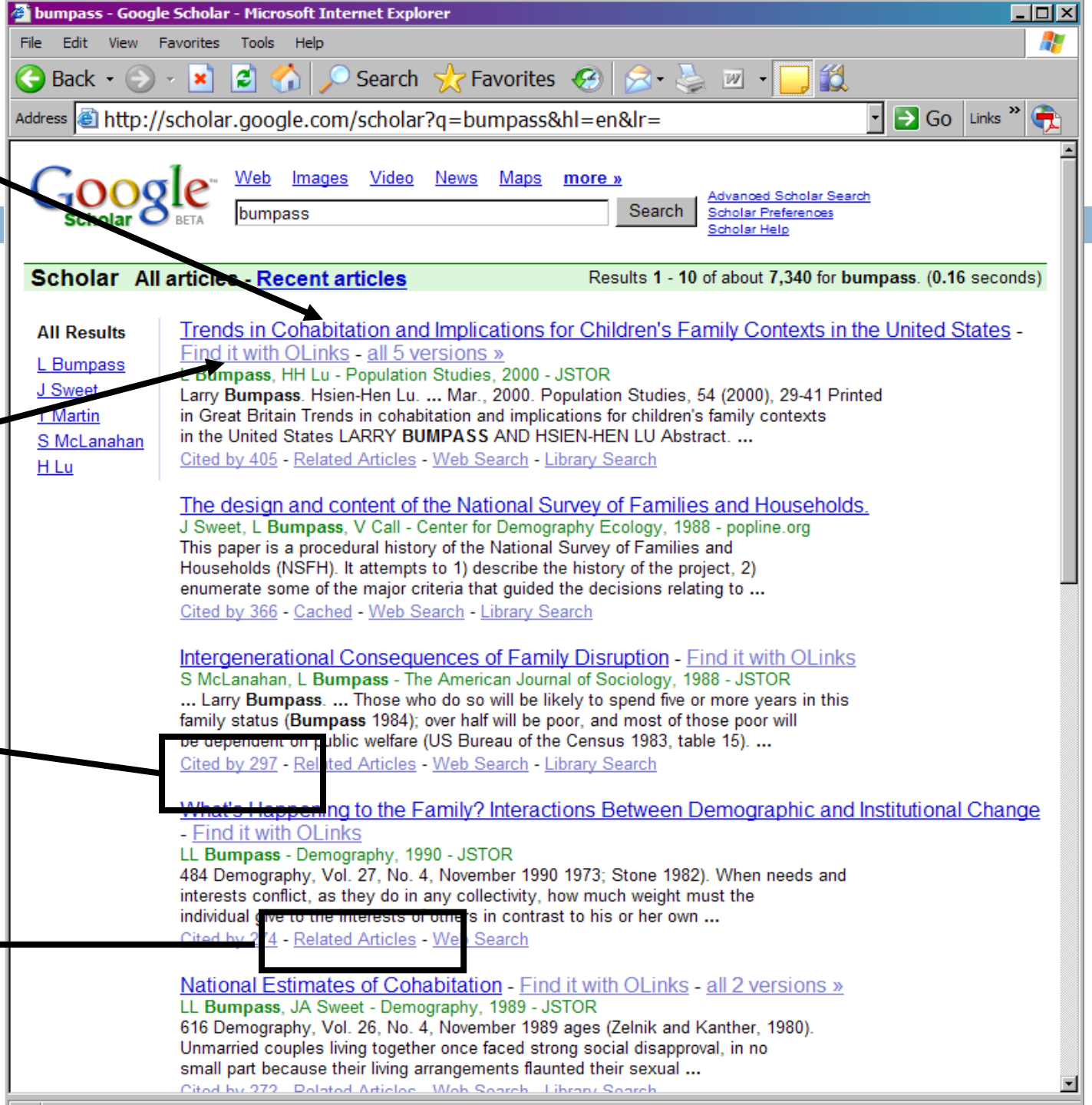
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Title links to abstract and possible source of full text

"Find it with OLinks" links to full text version

Link to articles that cited the work

Link to related articles



Google Scholar (search engine)

Advantages

- Searches journals, books and more academic sources.
- Can download individual citations into bibliographic managers.
- Contains citing information with links to sources citing a specific term.

Disadvantages

- No indexing terms.
- Huge retrieval of almost any topic.
- No ability to select citations for downloading or printing.
- Built in relevancy ranking based on times cited…
- May result in bias toward older literature.

PubMed Advanced Search Builder

PubMed®

User Guide

Add terms to the query box

All Fields | Enter a search term | ADD

Show Index

Query box

Enter / edit your search query here

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Search in: **Web of Science Core Collection** Editions: **All**

FIELDED SEARCH | QUERY BUILDER | CITED REFERENCES | STRUCTURE

Topic — Example: oil spill* mediterranean

And — All Fields — Example: liver disease india singh

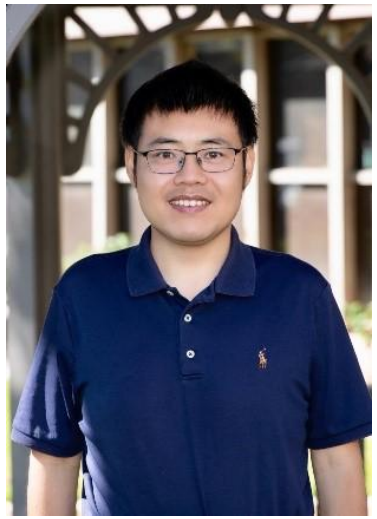
And — All Fields — Example: liver disease india singh

+ Add row + Add date range

Clear Search


Thank you!

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